

BOB-061 / BOB-062 â€” Suite stabilization + hang elimination

Revision: 1 **Last modified:** 2026-06-10T00:00:00Z **Status:** Fixed **Commits:** 6e15a8d, 6230865 **Mandate:** Â§11.4.50 (determinism), Â§11.4.98 (autonomous tests), Â§11.4.69/Â§11.4.102/Â§11.4.114.

BOB-061 â€” Unit-suite hang + order-dependent test-pollution

Root causes (proven via Â§11.4.102 systematic debugging + Â§11.4.114 clean-HEAD isolation):
1. **enricher full-suite hang** â€” `_lookup_tvmaze/_openlibrary/_musicbrainz` did real `aioshttp` GETs with no timeout; `musicbrainz` throttling stalled the event loop (the 87% monolithic stall). Fixed: `ClientTimeout(total=10)` on all 6 sessions + `test_resolve_no_apis` stubs the lookups (offline/autonomous). 2. **test-pollution** (13â€”34 order-dependent failures; `pytest-randomly` active): - `tests/conftest.py` `_CORRECT_MS_PATH` used `.parent` (=tests/) not `.parent.parent` (=repo root) â†’ a non-existent `tests/download-proxy/...` path corrupted `merge_service.__path__` â†’ 11 victims (all 6 `scheduler_api`, `TestLifespan`, `routes_coverage`). - raw `sys.modules` stubs (`helpers/env_loader/novaprinter/socks/tokyotoshokan/kinozal/rutor/iptorrents`) leaked across files â†’ extended `_POLLUTING_ROOTS`. - `socket.socket` (`kinozal SOCKS`) + `os.environ` (credential tests) leaked â†’ snapshot/restore per unit test in the autouse fixture.

BOB-062 â€” Unbounded loops + unbounded network I/O

- `kickass/bitsearch/torrentgalaxy while True:` search loops could run forever (index-ignoring upstream) â†’ `MAX_PAGES=50` caps + RED loop-cap regression tests.
- `search.py fetch_torrent`, `routes.py` `~3` qbit sessions, `helpers.py` `retrieve_url/download_file`, `eztv.py` fallback â†’ all given explicit timeouts (`aioshttp.ClientTimeout(10/30/urlopen timeout=30)`). Each with a RED-on-removal regression guard (mutation-verified by code review).
- `test_dashboard` qbit-auth test mocked (was a real `localhost:7185` connect).

Rock-solid proof (captured)

```
# monolithic â€” previously stalled, now completes:
pytest tests/unit/ -q --import-mode=importlib --timeout=60
â†’ 4121 passed, 0 failed in 324.56s
```

```
# determinism (Â§11.4.50) â€” top-level scope, multiple random seeds:
seed 7      â†’ 3150 passed, 0 failed
seed 42     â†’ 3150 passed, 0 failed
seed 100    â†’ 3150 passed, 0 failed
seed 31337  â†’ 3150 passed, 0 failed
seed 12345  â†’ 3150 passed, 0 failed
merge_service â†’ 801 passed  api_layer â†’ 170 passed
```

Clean-HEAD isolation (Â§11.4.114): HEAD 44f15b2 top-level = 34 failed/3070 passed; this work = 0 failed â†’ fixes are net-positive, introduce nothing. Two code-review passes returned GO (zero findings, zero warnings) after remediation (Â§11.4.134).